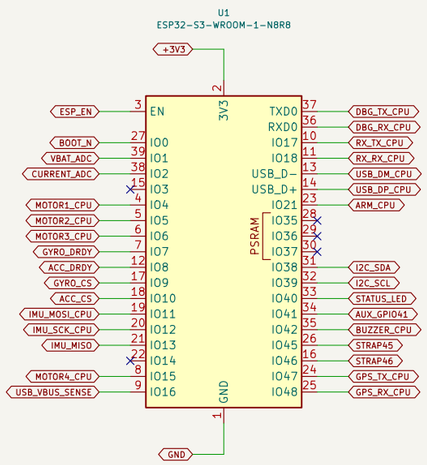
ESP32 FC / REV B

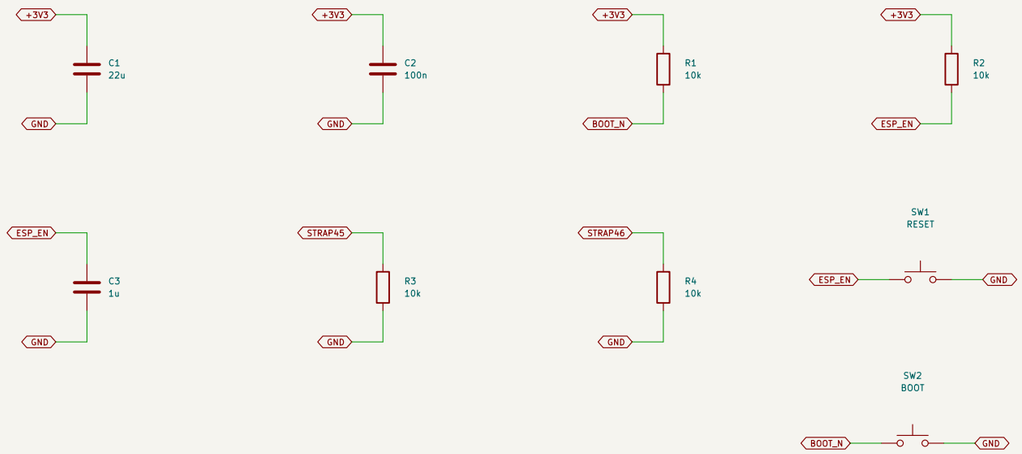
ENGINEERING SCHEMATIC – QUADCOPTER – EXTERNAL ESC

240 MHz dual-core controller



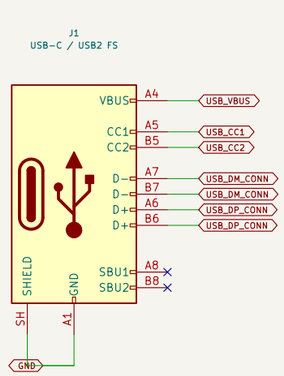
GPIO35/36/37 reserved for octal PSRAM.
GPIO3 unused; GPIO45/46 pulled low.
240 MHz CPU is a configuration target.
Keep antenna beyond PCB edge / copper keepout.

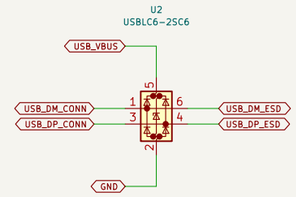
Module bypass / boot straps

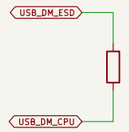


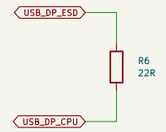
Hold BOOT, tap RESET, then release BOOT to download.
All motor outputs remain disabled until ARM_CPU goes high.

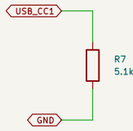
USB-C device / battery-powered programming

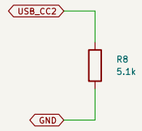


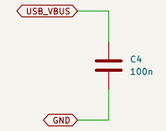


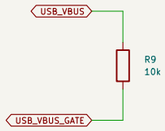


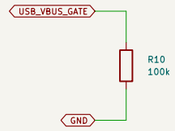


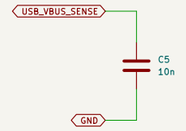


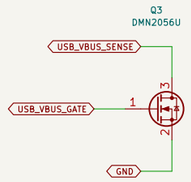


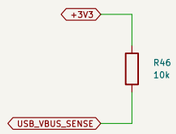












USB data only. Battery required. Q3 isolates VBUS sensing.
GPIO16 reads LOW when USB VBUS is present (Rev B).
GPIO19/20: USB full speed, 12 Mbps.
Firmware controls attach/detach. No USB charging.

Power

File: Power.kicad_sch

Sensors

File: Sensors.kicad_sch

Interfaces

File: Interfaces.kicad_sch

Auxiliary

File: Auxiliary.kicad_sch

Bench prototype; hardware and firmware untested.
3S–6S / External ESC / 3.3V logic
ESP32-S3 Flight Controller

Sheet: /
File: ESP32_FC.kicad_sch

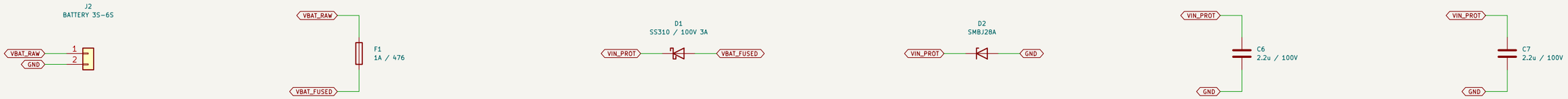
Title: 01 Processor and USB

Size: User	Date: 2026-10-15	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 1/5

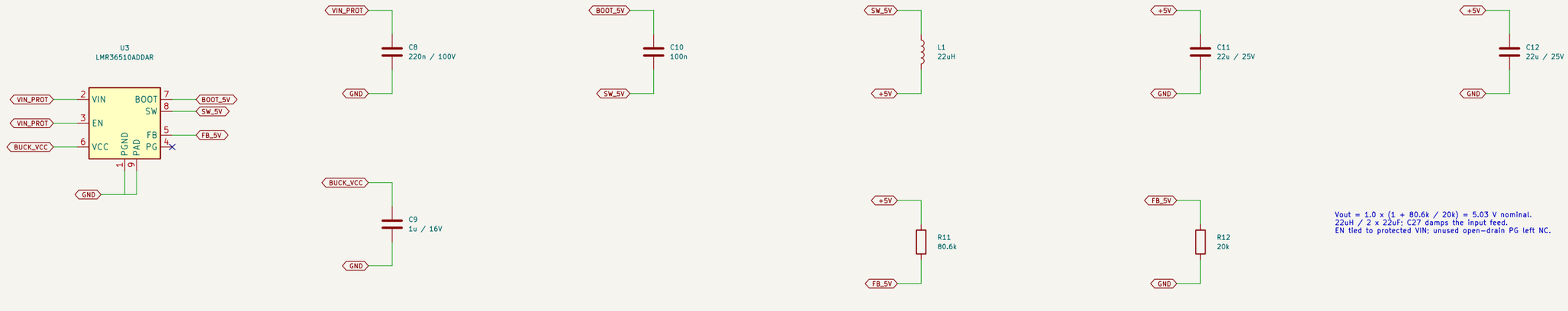
ESP32 FC / REV B

ENGINEERING SCHEMATIC – QUADCOPTER – EXTERNAL ESC

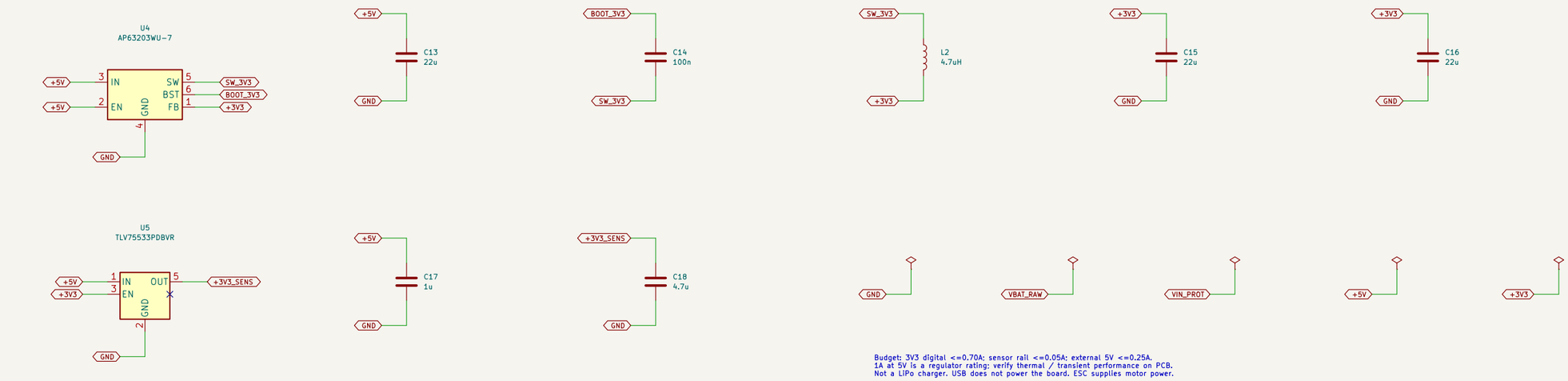
Battery entry / reverse-polarity protection



5 V / 1 A buck stage (shared system budget)



3.3 V digital buck / separate sensor LDO



Bench prototype; hardware and firmware untested.
3S–6S / External ESC / 3.3V logic

ESP32-S3 Flight Controller

Sheet: /Power/
File: Power.kicad_sch

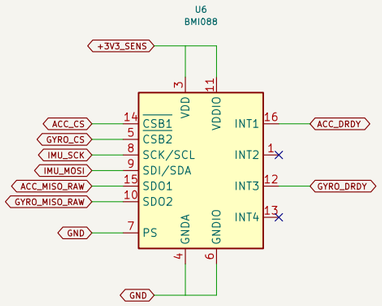
Title: 02 Battery and regulation

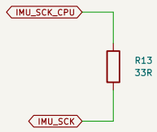
Size: User	Date: 2026-10-15	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 2/5

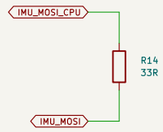
ESP32 FC / REV B

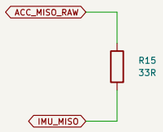
ENGINEERING SCHEMATIC – QUADCOPTER – EXTERNAL ESC

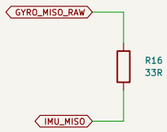
Dedicated SPI IMU / separate accelerometer and gyro CS

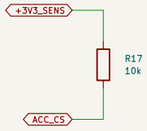


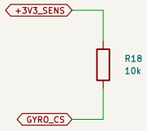


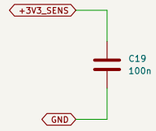


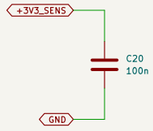


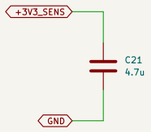




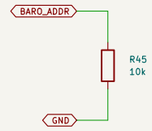




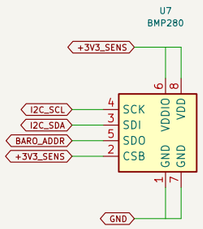


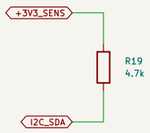


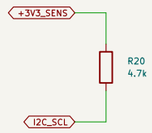
Start SPI at 1 MHz; qualify up to 10 MHz. Never select both dies at once.
Use gyro INT3 and accel INT1 for data-ready interrupts.
First accelerometer CS access selects SPI; follow Bosch startup and dummy-byte rules.
Keep IMU away from inductors, antenna, mounting stress and hot components.
Sensor axis orientation must be recorded during PCB layout.

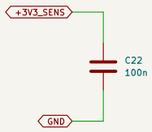


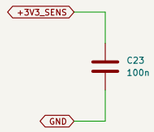
Barometric altitude / I2C expansion

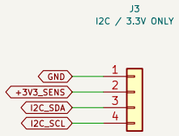




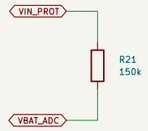


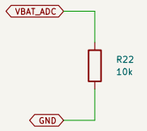


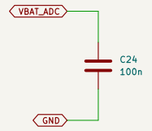


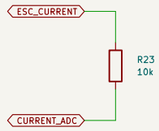


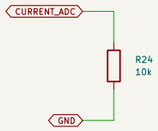
Battery voltage / ESC current monitoring

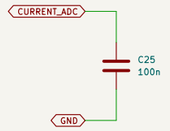












VBAT ADC = VIN_PROT / 16; includes the D1 voltage drop.
25.2V protected input -> 1.575V. GPIO1 / ADC1: calibrate offset. R21 = 150k.
Current ADC = ESC analog voltage / 2. GPIO2 / ADC1.
ESC current input: 0-5V only: calibrate V-to-A in firmware.
Do not connect digital telemetry or battery voltage to CURRENT.

Bench prototype; hardware and firmware untested.
3S-6S / External ESC / 3.3V logic
ESP32-S3 Flight Controller

Sheet: /Sensors/
File: Sensors.kicad_sch

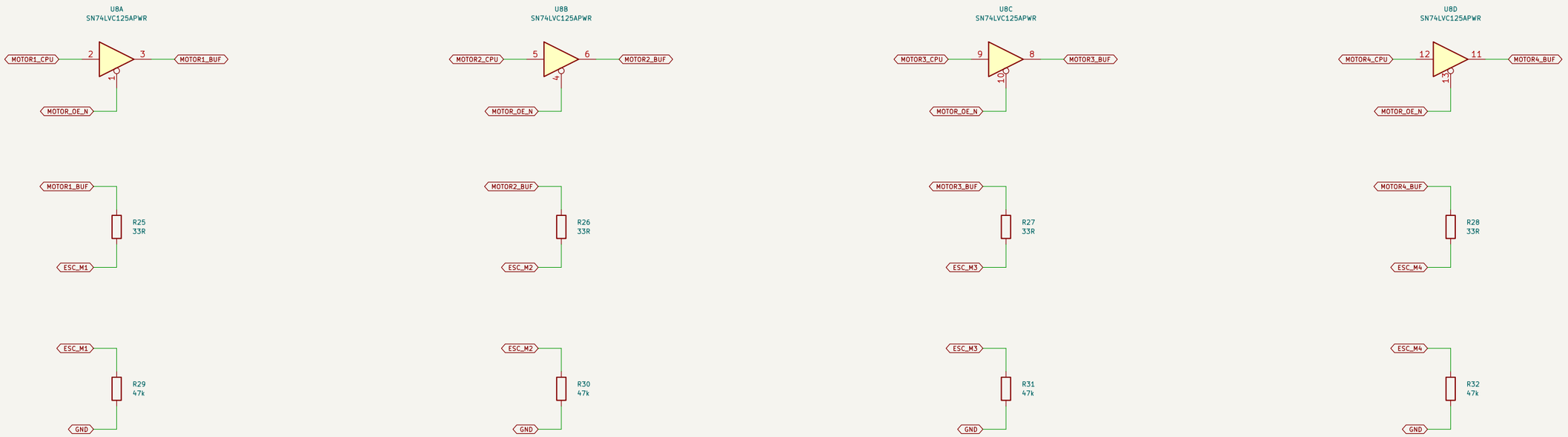
Title: 03 Sensors and monitoring

Size: User	Date: 2026-10-15	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 3/5

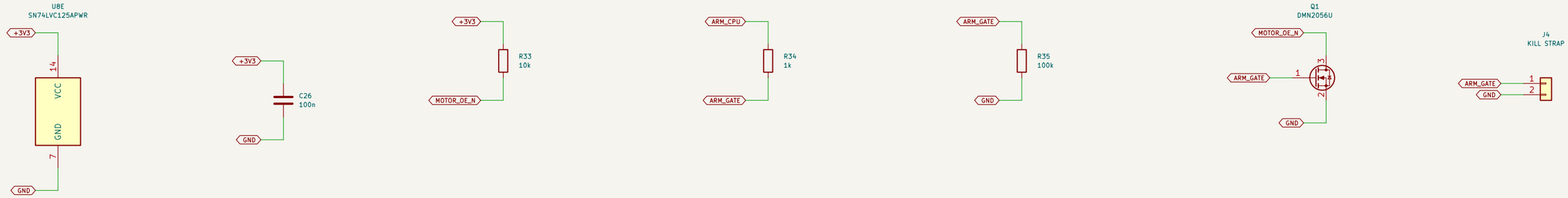
ESP32 FC / REV B

ENGINEERING SCHEMATIC – QUADCOPTER – EXTERNAL ESC

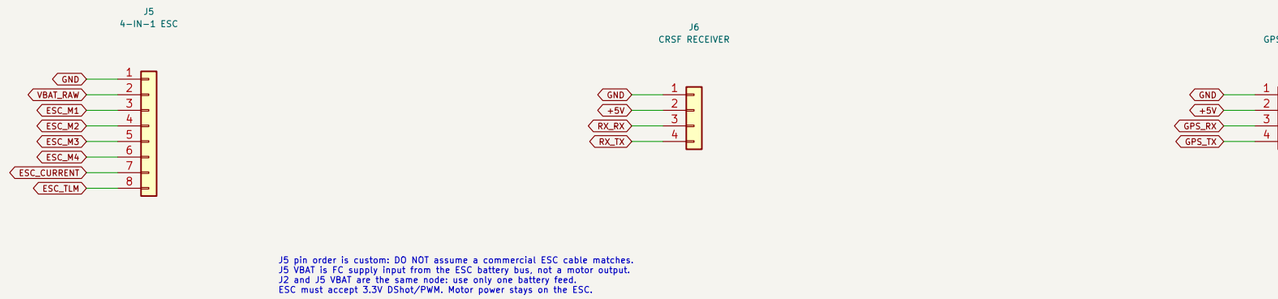
Four 3.3V motor outputs / hardware output-enable gate



ARM_CPU high -> Q1 on -> OE low -> motor signals enabled.
Reset / unpowered MCU -> Q1 off -> OE high -> outputs pulled low.
Firmware watchdog + receiver failsafe are still mandatory.
Unidirectional DShot/PWM only: no bidirectional DShot through U8.



External 4-in-1 ESC / receiver / GPS



J5 pin order is custom: DO NOT assume a commercial ESC cable matches.
J5 VBAT is FC supply input from the ESC battery bus, not a motor output.
J2 and J5 VBAT are the same node: use only one battery feed.
ESC must accept 3.3V DShot/PWM. Motor power stays on the ESC.

Bench prototype; hardware and firmware untested.
3S-6S / External ESC / 3.3V logic
ESP32-S3 Flight Controller

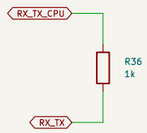
Sheet: /Interfaces/
File: Interfaces.kicad_sch

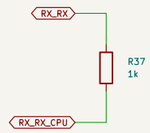
Title: 04 Motors and ESC interfaces

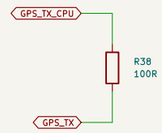
Size: User	Date: 2026-10-15	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 4/5

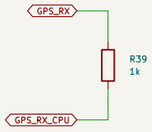
ESP32 FC / REV B

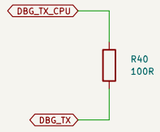
3.3V UART signal conditioning / target-side pin names

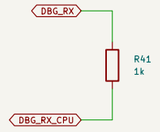




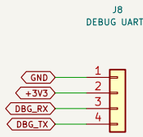


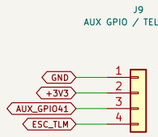




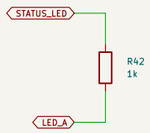


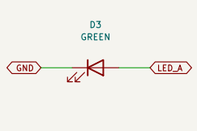
RX/TX labels are from the flight-controller perspective.
CRSF / ELRS: 3.3V non-inverted UART; receiver supply is 5V.
GPS must accept 5V supply and use 3.3V UART logic.
UART0 emits boot logs. GPS uses a separate remapped UART.
ESC_TLM is a breakout only; no MCU UART is allocated to it.

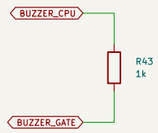


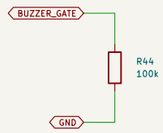


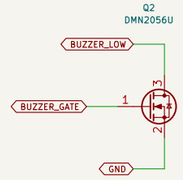
Status LED and switched external active buzzer

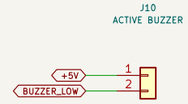














Bench prototype; hardware and firmware untested.
3S-6S / External ESC / 3.3V logic
ESP32-S3 Flight Controller

Sheet: /Auxiliary/
File: Auxiliary.kicad_sch

Title: 05 Serial ports and indicators

Size: User	Date: 2026-10-15	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 5/5